GAME DESIGN DOCUMENT

Wishing Reels

A 6×6 slot machine with gacha-inspired pity progression

Detail	Information
Genre	Slot Machine / Casual Probability-Based Game
Platform	Web Browser
Game Mode	Single-player
Perspective	2D Interface
Primary Inspiration	Genshin Impact's Wish and Pity System
Core Gameplay	6×6 Slot Machine with Gacha-Inspired Progression
Game Economy	Simulated GBP (£)
Author	Aashir Gurung
Document Purpose	Game Design Portfolio

1. Game Overview

1.1 Game Concept

Wishing Reels is a browser-based slot machine game that combines traditional slot machine mechanics with a gacha-inspired rarity and pity progression system.

The game takes inspiration from Genshin Impact's Wish system, particularly its soft pity, hard pity and featured reward guarantee mechanics. These systems have been adapted into a different gameplay genre to create a slot machine experience where random outcomes are balanced by predictable progression.

Players spend virtual currency to spin a 6×6 grid, matching symbols across horizontal, vertical and diagonal lines to receive rewards. Special symbols provide additional opportunities, including free spins and high-value jackpots.

The main distinguishing feature is the pity system. Every unsuccessful jackpot attempt increases the probability of receiving a 5★ reward, eventually guaranteeing one on the 90th spin.

Wishing Reels also introduces original mechanics that extend beyond the inspiration of Genshin Impact, including a carry-pot gambling decision, a safety net protection system, Pity Boost and gravity-based board animations.

> **In short:** The result is a game that combines slot machine gameplay, gacha-inspired progression and risk-versus-reward decision-making.

1.2 Game Vision

The vision for Wishing Reels is to demonstrate how an established progression mechanic from one genre can be adapted into another while retaining its underlying design principles.

Rather than relying entirely on random outcomes, the game gives players a visible sense of progression through its pity tracker.

The player is encouraged to continue playing through smaller line rewards, occasional free spins and the expectation of an eventual jackpot.

The additional carry-pot system introduces a layer of player choice, allowing players to decide whether to secure their winnings or risk them for a potentially larger reward.

1.3 Design Pillars

#	Pillar	Description
1	Randomness with Predictable Progression	The game uses random reward generation but limits the maximum number of spins required to receive a 5★ jackpot.
2	Meaningful Rewards	Different reward rarities provide different levels of value and gameplay opportunities.
3	Player Choice and Risk Management	The carry-pot and safety net mechanics allow players to make decisions that influence their potential rewards and losses.
4	Transparent Progression	The pity tracker, reward history and statistics make the progression system understandable and visible.

#	Pillar	Description
5	Original Mechanical Adaptation	The game uses Genshin Impact's pity system as a design reference while introducing its own slot-based gameplay, reward structure and additional mechanics.

2. Design Research and Originality

2.1 Primary Inspiration: Genshin Impact

The main research reference for Wishing Reels is Genshin Impact's Wish system.

Genshin Impact uses a gacha system in which players obtain characters and weapons through randomised wishes. Its reward structure includes rarity tiers, soft pity, hard pity and featured reward guarantees.

These mechanics were particularly interesting from a game design perspective because they introduce progression into an otherwise random reward system.

The pity system addresses one of the main weaknesses of low-probability reward systems: the possibility that players can experience extended periods without receiving a desirable reward.

By gradually increasing the probability of a high-rarity reward and introducing a guaranteed threshold, the system creates a more predictable progression experience.

2.2 Adaptation into Wishing Reels

Rather than recreating Genshin Impact's Wish system directly, Wishing Reels adapts its underlying principles into a different gameplay structure.

Genshin Impact	Wishing Reels
Character and weapon wishes	6×6 slot machine spins
Primogems and Fates	Simulated GBP
3★, 4★ and 5★ rewards	3★, 4★ and 5★ rewards
Character soft pity	Jackpot soft pity
90-pull hard pity	90-spin hard pity
Featured character 50/50	Featured jackpot 50/50
Guaranteed featured character	Guaranteed featured jackpot
Character banner rewards	Slot-based reward symbols

The main design principle retained from Genshin Impact is the use of pity as a progression system that reduces the unpredictability of random rewards.

However, several elements of Wishing Reels are original adaptations rather than features taken from Genshin Impact.

2.3 Original Mechanics

The following mechanics were designed specifically for Wishing Reels:

Mechanic	Description
6×6 Symbol Grid	The game uses a six-column, six-row board where players can receive rewards through matching symbols.
Line-Based Rewards	Players can receive smaller payouts by matching three, four or five identical symbols across different directions.
Free Spin System	The 4★ reward triggers two additional spins, with the second free spin receiving a higher line-win multiplier.
Pity Boost	Players can spend virtual currency to increase their pity counter by ten.
Carry-Pot System	Players can choose to carry a featured jackpot into another attempt, potentially doubling their winnings.
Safety Net	Players can pay a percentage of their carried jackpot to retain part of it if the next jackpot is standard.
Gravity Mode and Board Events	The game includes additional visual features that make the slot machine feel more dynamic.

These original systems are what distinguish Wishing Reels from a direct recreation of Genshin Impact's gacha system.

3. Target Audience

Wishing Reels is designed for:

- Players interested in slot machine games.
- Players familiar with gacha mechanics.
- Genshin Impact players interested in pity and rarity systems.
- Casual players looking for short gameplay sessions.
- Players who enjoy probability-based progression and reward management.

> **Important:** The game uses fictional currency rather than real-money gambling. All monetary values are simulated GBP and cannot be withdrawn or exchanged for real money.

4. Core Gameplay Loop

4.1 Gameplay Flow

1. The player starts with £200 in virtual currency.
2. The player spends £2 to initiate a paid spin.
3. The game determines the spin's reward outcome.
4. A 6×6 grid is generated to visually represent that outcome.
5. The game checks for matching lines and special rewards.
6. Rewards are calculated and added to the player's balance.
7. The pity counters are updated.
8. The player receives additional free spins if a 4★ reward is triggered.

9. If a featured jackpot is obtained, the player can choose to cash out or carry the winnings.
10. The player continues playing until they stop or run out of virtual currency.

4.2 Core Gameplay Loop

Balance → Spin → Determine Outcome → Generate Grid → Display Rewards → Update Pity → Repeat

The game is designed around a repeatable gameplay loop, with the pity tracker providing longer-term progression between individual spins.

5. Slot Machine Mechanics

5.1 Grid Structure

Wishing Reels uses a 6×6 slot grid consisting of 36 individual cells.

Each cell displays a symbol generated through the game's symbol-generation system. The board contains:

- Common symbols.
- Free-spin symbols.
- Featured jackpot symbols.
- Standard jackpot symbols.

The grid provides the main visual representation of the game's reward outcomes.

5.2 Symbol Types

Symbol	Rarity	Function
🗡️	3★	Common matching symbol
🛡️	3★	Common matching symbol
🏹	3★	Common matching symbol
📖	3★	Common matching symbol
🪙	3★	Common matching symbol
🧪	3★	Common matching symbol
🔮	4★	Free-spin reward
👑	5★ Featured	Main jackpot
🗝️	5★ Standard	Secondary jackpot

5.3 Symbol Matching

Players can receive smaller rewards by matching three, four or five identical common symbols in a straight line. Valid matching directions include:

- Horizontal.
- Vertical.
- Diagonal downwards.

- Diagonal upwards.

The matching reward is determined by the length of the line.

Matching Symbols	Reward
3 symbols	£1
4 symbols	£3
5 symbols	£6

The current implementation generates one intended matching line for an ordinary line-win outcome.

5.4 Line-Win Outcome Distribution

The current implementation uses the following probabilities when generating ordinary line-win outcomes:

Outcome	Probability
5-symbol line	2%
4-symbol line	7%
3-symbol line	21%
No line	70%

These probabilities are applied when the spin produces an ordinary 3★ outcome.

6. Rarity and Reward System

6.1 3★ Rewards – Common Rewards

3★ outcomes represent the most common results. Players can receive:

- No matching line.
- Three-symbol matching lines.
- Four-symbol matching lines.
- Five-symbol matching lines.

These outcomes provide smaller payouts and maintain engagement between higher-rarity rewards.

6.2 4★ Rewards – Free Spins

The 4★ reward is represented by the 🔮 symbol. When a 4★ outcome is obtained during a paid spin, the player receives two free spins.

Free Spin	Line-Win Multiplier
First free spin	×1
Second free spin	×2

- Free spins do not require an additional £2 payment.

- However, they still contribute towards both pity counters.
- A 4★ result during a free spin does not trigger another free-spin sequence.
- If a 5★ reward appears during a free spin, the remaining free-spin sequence ends.
- The 4★ pity counter guarantees at least one 4★ result within ten pulls, subject to the 5★ outcome taking priority when both thresholds are reached on the same pull.

6.3 5★ Rewards – Jackpot

The 5★ reward is the highest rarity tier. There are two possible jackpot outcomes:

Symbol	Reward	Description
👑	£80	Featured jackpot
🗝️	£26	Standard jackpot

The featured jackpot is the main prize, while the standard jackpot provides a smaller reward. A 5★ result resets the 5★ pity counter.

7. Gacha-Inspired Pity System

7.1 System Overview

The pity system is the main progression mechanic in Wishing Reels. It tracks the number of spins made since the previous 5★ jackpot. Every unsuccessful 5★ attempt increases the player's progress towards the next jackpot.

The system contains two main thresholds:

- Soft Pity: The point where the 5★ probability begins increasing.
- Hard Pity: The point where a 5★ reward is guaranteed.

7.2 Base Probability

The initial 5★ probability is 0.6% per pull. From pulls 1 to 73, the probability remains at 0.6%. Starting at pull 74, the probability increases by approximately six percentage points per pull.

7.3 Soft Pity

Soft pity begins at pull 74.

Pull Number	5★ Probability	Pull Number	5★ Probability
1–73	0.6%	82	54.6%
74	6.6%	83	60.6%
75	12.6%	84	66.6%
76	18.6%	85	72.6%
77	24.6%	86	78.6%
78	30.6%	87	84.6%

Pull Number	5★ Probability	Pull Number	5★ Probability
79	36.6%	88	90.6%
80	42.6%	89	96.6%
81	**48.6%**	**90**	**100%**

The purpose of soft pity is to make the likelihood of receiving a jackpot increase as the player approaches hard pity.

7.4 Hard Pity

Hard pity guarantees a 5★ jackpot on the 90th pull if the player has not received one earlier. Once a 5★ reward is obtained, the 5★ pity counter resets to zero.

This ensures that players cannot continue indefinitely without receiving a jackpot.

7.5 Pity Tracker

The pity tracker is displayed alongside the slot machine. It provides:

- Current spins since the previous jackpot.
- Current 5★ probability.
- Remaining pulls until hard pity.
- Soft pity threshold.
- Hard pity threshold.
- Next jackpot guarantee status.

The visual progress meter fills as the player approaches the 90-pull guarantee.

8. Featured Jackpot and 50/50 System

8.1 Overview

When a player obtains a 5★ reward, the game determines whether the result is the featured jackpot or the standard jackpot. The initial probability is:

- 50% chance of receiving 👑 Featured.
- 50% chance of receiving 🗝️ Standard.

8.2 Winning the 50/50

If the player receives the featured 👑 jackpot, the 50/50 is won. The player receives £80. The next jackpot returns to the standard 50/50 system unless another mechanic changes the reward state.

8.3 Losing the 50/50

If the player receives the standard 🗝️ jackpot, the 50/50 is lost. The player receives £26, and their next 5★ reward is guaranteed to be the featured 👑 jackpot. This guarantee remains active until the player receives the featured jackpot.

8.4 Guarantee Behaviour

Previous Result	Next 5★ Result
No active guarantee	50% Featured / 50% Standard
Lost 50/50	100% Featured
Won featured jackpot	50% Featured / 50% Standard

The guarantee provides an additional progression layer on top of the pity system.

9. Signature Mechanic: Carry-Pot System

9.1 Design Overview

The carry-pot system is one of the main original mechanics in Wishing Reels. It introduces a risk-versus-reward decision after obtaining a featured jackpot.

Rather than automatically collecting the £80 reward, players can choose to carry their winnings into another jackpot attempt. This changes the player's immediate reward from a secured payout into a potential larger payout.

Unlike the pity system, which controls reward probability, the carry-pot system changes how a jackpot reward is managed after it has been obtained.

9.2 Cash Out

The player can choose to collect the featured jackpot immediately. The full reward is transferred into their virtual currency balance. This is the lower-risk option because the player secures their winnings without risking the jackpot again.

9.3 Carry

If the player chooses to carry the reward, the winnings become a carry pot. The game immediately sets the 5★ pity counter to 73.

This is a deliberate design decision. Instead of resetting pity to zero after the featured jackpot, the carry mechanic places the player at the beginning of soft pity. The next spin therefore has a 6.6% chance of producing a 5★ reward. If the player does not receive a jackpot, the pity probability continues increasing normally.

The next 5★ result determines what happens to the carried winnings.

Next Jackpot	Result
👑 Featured	Carry pot doubles
🗝️ Standard	Carry pot is lost
🗝️ Standard with safety net	Player retains 50% of the original carry pot

9.4 Carry-Pot Example

Assume the player receives a featured jackpot worth £80. Instead of cashing out, they choose to carry. Their carry pot is now £80, and their pity counter becomes 73. The next spin has a 6.6% 5★ probability. If the player does not receive a jackpot, the pity counter continues increasing. Eventually, when a 5★ reward is obtained:

Scenario A – Featured Jackpot

- The carry pot doubles.
- £80 becomes £160.
- The player receives another cash-out or carry decision.

Scenario B – Standard Jackpot

- The carry pot is lost.
- The player receives no carry-pot payout.
- Their next 5★ is guaranteed to be featured.
- If the safety net was purchased, the player instead retains half of the original carry pot.

9.5 Why Pity Starts at 73

The decision to set pity to 73 is an important part of the carry-pot balance. It allows players to continue from just before soft pity rather than starting a completely new pity cycle.

This provides a meaningful advantage compared with an ordinary new jackpot attempt. However, it does not guarantee an immediate jackpot. The player still needs to progress through the increasing probability thresholds, and the 50/50 system remains relevant.

This creates a trade-off:

- Cashing out secures the current reward.
- Carrying creates an opportunity to double the reward.
- The pity jump reduces the distance to the next jackpot.
- Losing the next 50/50 risks the entire carried amount.

The mechanic is designed to make jackpot management an additional layer of gameplay rather than simply an automatic reward collection process.

9.6 Carry-Pot Balance Considerations

The carry-pot mechanic needs particular attention during balancing because it changes the relationship between reward value and jackpot probability.

The initial 6.6% chance after carrying should not be interpreted as a guaranteed or immediate second jackpot. Players may require multiple spins before reaching another 5★ reward.

The main balancing variables are:

- Starting carry-pot value.
- Pity starting point of 73.
- Featured versus standard probability.
- Doubling multiplier.

- Safety net cost.
- Safety net protection percentage.
- Cost of subsequent spins.

> **Note:** These variables should be tested together rather than balanced independently.

10. Safety Net System

10.1 Overview

The safety net is an optional protection feature within the carry-pot system. It gives players an opportunity to reduce their potential loss when choosing to carry a jackpot.

10.2 Activation

When choosing to carry, the player can activate the safety net by paying 25% of the current carry-pot value. The cost is deducted immediately from the player's balance.

10.3 Protection

If the next jackpot is standard, the player retains 50% of the original carry pot. If the next jackpot is featured, the carry pot doubles normally. The safety net only applies to the current carry attempt.

10.4 Example

Assume the player has an £80 carry pot.

Feature	Amount
Current carry pot	£80
Safety net cost	£20
Amount retained after a standard jackpot	£40
Amount after a featured jackpot	£160

The safety net reduces the potential loss but comes at an additional cost.

> **Note:** It should not be described as mathematically neutral or as having no effect on expected returns. Its actual impact must be measured through probability and economy testing.

10.5 Design Purpose

The safety net introduces an additional decision. Players must consider whether paying a percentage of their winnings is worthwhile in exchange for partial protection. This gives the carry-pot system more strategic depth and creates a choice between maximum potential reward and reduced potential loss.

11. Free Spin System

11.1 Trigger Condition

The player receives two free spins whenever a 4★ reward is obtained during a paid spin.

11.2 Free Spin Rewards

Outcome	First Free Spin	Second Free Spin
3-symbol line	£1	£2
4-symbol line	£3	£6
5-symbol line	£6	£12

11.3 Free Spin Restrictions

- Free spins cannot retrigger additional free spins.
- A 5★ result ends the remaining free-spin sequence.
- Free spins still advance both pity counters.
- The second free spin has a higher line-win multiplier.
- Free spins do not require additional currency.

The system provides additional reward opportunities without requiring the player to spend more virtual currency.

12. Pity Boost System

12.1 Overview

The Pity Boost allows players to spend virtual currency to accelerate their progress towards a 5★ reward. It provides an alternative to waiting for the natural pity counter to increase.

12.2 Cost and Effect

Feature	Value
Cost	£12
Pity Increase	+10
Maximum Pity	89
Usage	Repeatable when eligible

Each activation increases the pity counter by ten, up to a maximum of 89. The player cannot use the boost while a spin is processing or while a jackpot decision is pending.

12.3 Design Purpose

The Pity Boost provides a currency-for-progression exchange. Players can spend additional virtual currency to move closer to soft pity or hard pity.

It is an optional shortcut rather than a necessary gameplay requirement. Its effect on the overall economy must be included in balancing tests.

13. Virtual Currency and Economy

13.1 Currency Overview

Wishing Reels uses a fictional GBP-based economy. All displayed monetary values are simulated and have no real-world monetary value.

13.2 Starting Balance

Every new game begins with £200.

13.3 Spin Cost

Each paid spin costs £2. Free spins do not require additional payment.

13.4 Top-Up

Players can obtain additional virtual currency through the top-up button. Each activation adds £100 to the player's balance. This is a simulated currency feature and does not involve real-money transactions.

13.5 Economy Summary

Feature	Cost / Value
Starting Balance	£200
Paid Spin	£2
Top-Up	£100
Pity Boost	£12
Featured Jackpot	£80
Standard Jackpot	£26
Three-Symbol Line	£1
Four-Symbol Line	£3
Five-Symbol Line	£6

14. Additional Gameplay Features

14.1 Gravity Mode

Gravity mode is an optional visual feature. When enabled, winning symbols burst and disappear from the board. New symbols then fall into their positions, creating a more dynamic slot-machine presentation. When disabled, the board remains static after a winning result.

Gravity mode does not change the underlying reward probabilities.

14.2 Turbo Mode

Turbo mode reduces animation duration. It allows players to progress through spins more quickly without changing probability or reward calculations.

14.3 Auto-Spin

Auto-spin automatically initiates consecutive spins. It continues until:

- The player manually stops auto-spin.
- The player runs out of virtual currency.
- A featured jackpot requires a cash-out or carry decision.

14.4 Board Events

Random visual events can occur during a spin.

Event	Effect
Mirror Event	The board flips horizontally, creating a mirrored visual effect.
Lightning Event	Nine randomly selected cells briefly flash with a lightning-style visual effect.

These events are cosmetic and do not affect the player's odds or rewards.

15. User Interface and User Experience

15.1 Main Screen Layout

The main interface is divided into two primary sections.

Slot Machine Area

- Game title.
- 6×6 slot grid.
- Current win announcement.
- Spin button.
- Auto-spin button.
- Pity Boost button.
- Gravity mode toggle.
- Turbo mode toggle.
- Virtual currency balance.
- Top-up button.

Pity Tracker and Statistics Area

- Pity progress meter.
- Current 5★ probability.
- Current pity count.
- 4★ guarantee countdown.
- Featured guarantee status.
- Maximum remaining paid spin cost to reach hard pity.
- Carry-pot value.
- Total spins.

- Total jackpots.
- Featured jackpots.
- Average spins per jackpot.
- Net winnings.
- Recent reward history.

15.2 Visual Feedback

The game uses visual effects to communicate reward rarity.

Rarity	Presentation
3★ Rewards	Standard board presentation with highlighted matching lines.
4★ Rewards	Violet visual highlighting and free-spin notifications.
5★ Rewards	Gold visual highlighting and jackpot announcements.

15.3 Colour Palette

Swatch	Colour	Purpose
	Gold	Featured jackpots and high-value rewards
	Violet	4★ rewards
	Jade	Progression and pity meter
	Navy	Main interface and buttons
	Cream	Slot grid background

15.4 Responsive Design

The interface supports desktop and mobile browser layouts. On desktop, the slot machine and pity tracker are displayed side by side. On smaller screens, the layout switches to a single-column arrangement.

16. Player Statistics and Reward History

16.1 Statistics

The game records:

- Total spins.
- Total 5★ rewards.
- Total featured jackpots.
- Average number of spins per jackpot.
- Current pity.
- Current 4★ pity.
- Current featured guarantee status.

- Current carry-pot value.
- Total winnings.
- Total currency spent.
- Net winnings or losses.

16.2 Reward History

The reward history records recent 4★ and 5★ results. For 5★ rewards, the history displays:

- Reward rarity.
- Featured or standard result.
- Total spin number.
- Pity count at the time of the reward.

This provides transparency and allows players to review their recent results.

17. Technical Design and Implementation

17.1 Technology

The current implementation uses:

Technology	Purpose
HTML	Game structure
CSS	Styling, layout and animations
JavaScript	Gameplay logic and randomisation
Browser Local Storage	Saving player progress

17.2 Reward Generation

The current implementation determines the reward outcome before generating the final visual arrangement of the board. The game first evaluates the pity system and determines whether the result is 3★, 4★ or 5★. For ordinary 3★ outcomes, it determines whether a matching line is generated. The board is then populated to visually represent the selected outcome.

This means the current game is outcome-driven rather than a system where the 36 independently randomised cells determine the reward.

> **Note:** This is an intentional implementation detail that separates reward probability from board presentation.

17.3 Random Number Generation

The game currently uses JavaScript's Math.random() function to generate random outcomes. Randomisation is used for:

- Initial symbol placement.
- Line-win outcomes.

- 4★ and 5★ reward checks.
- Featured versus standard jackpot selection.
- Board events.

The pity system modifies the probability of receiving a 5★ reward based on the current pity counter.

17.4 Pity Logic

The game maintains persistent variables for:

- 5★ pity counter.
- 4★ pity counter.
- Featured guarantee status.
- Total spins.
- Total 5★ rewards.
- Total featured rewards.
- Carry-pot value.
- Safety net status.

The 5★ pity counter resets when a jackpot is obtained. The 4★ pity counter resets when a 4★ reward is obtained.

17.5 Save System

The current game uses browser Local Storage to save player progress. Saved information includes:

- Virtual currency balance.
- Pity counters.
- Featured guarantee status.
- Spin statistics.
- Reward history.
- Carry-pot information.
- Total currency spent and won.

Players can reset their progress using the reset button.

17.6 Animation System

The game uses CSS animations and JavaScript timing to create the spinning effect. Each column animates independently before revealing its final symbols. Additional animations include:

- Winning symbol highlighting.
- Symbol bursting.
- Gravity falling.
- Board flipping.
- Lightning effects.
- Jackpot notifications.

The game also respects reduced-motion preferences where supported.

18. Economy Balancing and RTP Validation

18.1 Current Economy Status

The current implementation has an intended long-term payback target of approximately 94%. However, this figure has not yet been independently verified through statistical simulation.

> **Important:** It should therefore be treated as a provisional design target rather than a confirmed property of the game.

This distinction is important because the pity system, free spins, Pity Boost and carry-pot mechanics all influence the overall economy. The actual return depends on how frequently each reward occurs and how players interact with the optional mechanics.

18.2 Understanding Return to Player

Return to Player (RTP) measures how much of the currency spent by players is returned through rewards over a large number of gameplay attempts. For Wishing Reels, the basic calculation is:

RTP = (Total Rewards Paid ÷ Total Paid Spin Stakes) × 100

For example, if players collectively spend £10,000 on paid spins and receive £9,400 in rewards, the RTP is 94%. This is a long-term statistical measure rather than a guarantee of what an individual player will receive.

18.3 Why the Current RTP Needs Testing

Several mechanics affect the overall return:

Mechanic	Effect on Return
Pity System	The 5★ probability increases over time, meaning the average jackpot frequency is higher than the base 0.6% probability suggests.
4★ Free Spins	Free spins provide additional opportunities to earn line rewards and potentially receive a 5★ jackpot without another paid stake.
Pity Boost	The Pity Boost allows players to spend additional currency to advance their pity counter.
Carry-Pot System	Carrying a jackpot creates an opportunity to double the reward, but also introduces the possibility of losing the carried pot.
Safety Net	The safety net adds an additional cost while reducing the potential loss of a carried jackpot.

These systems must be evaluated together to determine the actual long-term return.

18.4 RTP Testing Methodology

A Monte Carlo simulation should be used to test the economy. The simulation will run a large number of automated gameplay sessions using the same reward probabilities and mechanics as the actual game.

Suggested testing process:

1. Simulate a large number of paid spins.
2. Apply the 5★ soft-pity and hard-pity probabilities.

3. Apply the 4★ guarantee and free-spin mechanics.
4. Include line-win probabilities and reward multipliers.
5. Track featured and standard jackpot outcomes.
6. Simulate carry-pot decisions.
7. Include safety net costs and protection.
8. Test Pity Boost usage separately and as part of different player strategies.
9. Calculate total currency spent and total rewards paid.
10. Compare the resulting RTP across different player strategies.

18.5 Testing Scenarios

The following scenarios should be evaluated independently.

Test	Player Behaviour	Purpose
A	Always cashes out	Establish baseline economy
B	Always carries featured jackpots	Evaluate carry-pot risk
C	Always uses safety net	Measure protection cost
D	Never uses Pity Boost	Establish natural pity performance
E	Uses Pity Boost whenever possible	Measure acceleration cost
F	Uses mixed strategies	Evaluate typical gameplay behaviour

18.6 Metrics to Record

The simulation should record:

- Total paid spins.
- Total virtual currency spent on paid spins.
- Total rewards paid.
- Total 4★ rewards.
- Total 5★ rewards.
- Average spins per jackpot.
- Average jackpot payout.
- Number of featured jackpots.
- Number of standard jackpots.
- Total free spins generated.
- Total line rewards.
- Total carry-pot wins.
- Total carry-pot losses.
- Total safety net costs.
- Total Pity Boost costs.
- Overall RTP.

18.7 Balancing Target

The provisional economy target is approximately 94% RTP under a clearly defined baseline strategy. This target should only be confirmed after simulation.

If the results differ significantly, the following variables can be adjusted:

- Common line-win probabilities.
- Line-win payouts.
- 4★ reward probability.
- 5★ pity acceleration.
- Featured and standard jackpot payouts.
- Carry-pot doubling rules.
- Safety net cost and protection percentage.
- Pity Boost cost.

The objective is to create a balanced virtual economy while retaining the intended excitement of the pity and carry-pot systems.

19. Known Implementation Considerations

The following points are important when maintaining or expanding the current version.

Area	Consideration
Pity Boost	The Pity Boost increases the counter by ten but does not directly guarantee a jackpot. It can bring the player to 89, with the next pull reaching hard pity.
Carry-Pot Behaviour	The carry-pot system is activated following a featured jackpot. A standard jackpot while carrying resolves the existing pot and does not create another carry decision.
Featured Guarantee	Losing the 50/50 activates a featured guarantee. This guarantee remains active even if the carried pot is lost.
Free Spins	Free spins advance the pity counters and can produce 5★ rewards. However, a 4★ reward during a free spin does not trigger another free-spin sequence.
Board Generation	The current implementation selects reward outcomes before generating the final board. The visual grid represents the selected outcome rather than independently determining it.
Board Events	Mirror and lightning events are visual effects. They do not influence reward probabilities.
Persistence	The current save system is browser-based. Progress is stored locally rather than through a central account or server.
Economy Validation	The approximately 94% payback figure remains a provisional target until it has been validated through simulation.

20. Future Development Opportunities

The current version establishes the main gameplay systems. Future development could expand the game through the following features.

20.1 Additional Slot Themes

Introduce different visual themes and symbol sets to provide variety without changing the core pity system.

20.2 Additional Reward Tiers

Introduce additional rarity levels, potentially with different payout values and special effects.

20.3 Expanded Jackpot Features

Introduce additional jackpot choices or alternative carry-pot challenges.

20.4 Improved Statistics

Add:

- Long-term win-rate tracking.
- Highest jackpot achieved.
- Longest losing streak.
- Total free spins received.
- Total money earned through line matches.
- Historical pity distribution.

20.5 Sound Design

Add audio feedback for:

- Regular spins.
- Winning lines.
- Free-spin triggers.
- Soft pity activation.
- Featured jackpots.
- Standard jackpots.
- Carry-pot decisions.

20.6 Probability and Balance Testing

Develop a dedicated simulation tool to test large numbers of spins and measure actual return-to-player percentages. This would help ensure that the virtual economy remains balanced across different player strategies.

20.7 Improved Save Architecture

Consider implementing a more robust save system if the game is expanded beyond its current browser-based prototype.

21. Glossary

Term	Definition
3★	Common reward tier
4★	Reward tier that triggers two free spins
5★	Highest reward tier, representing a jackpot
Featured Jackpot	The £80 👑 reward
Standard Jackpot	The £26 🗝️ reward
Pity	Counter tracking pulls since the previous 5★
Soft Pity	Pull 74, where jackpot probability starts increasing
Hard Pity	Pull 90, where a jackpot is guaranteed
50/50	Initial 50% chance of receiving the featured jackpot
Guarantee	Next 5★ is guaranteed featured after losing the 50/50
Free Spin	A spin that does not require additional payment
Pity Boost	Paid feature that adds ten to the pity counter
Carry Pot	A featured jackpot reward held for another attempt
Safety Net	Protection that retains half the carry pot after a standard jackpot
Gravity Mode	Visual effect where winning symbols burst and new symbols fall
Turbo Mode	Faster spin animations
RTP	Long-term proportion of currency spent that is returned as rewards
Monte Carlo Simulation	A statistical method using repeated randomised trials to estimate system behaviour

22. Conclusion

Wishing Reels demonstrates how an established gacha progression mechanic can be adapted into a different gameplay genre.

The game takes inspiration from Genshin Impact's pity system, using soft pity, hard pity and featured reward guarantees to create a more predictable reward progression experience.

However, the project extends beyond this inspiration through its original 6×6 slot grid, line-based rewards, free-spin system, Pity Boost and carry-pot mechanic.

The carry-pot system is a particularly important part of the game’s identity because it introduces an additional layer of player decision-making. By allowing players to carry a featured jackpot, start at pity 73 and potentially double their winnings, the game creates a risk-versus-reward mechanic that complements the underlying pity system.

The project also demonstrates an awareness of economy design and probability balancing. Rather than treating the intended 94% RTP as an established result, the design identifies statistical simulation as a necessary next step for validating the game's reward distribution.

Ultimately, Wishing Reels is an exploration of how mechanics from an existing game system can be researched, adapted and expanded into an original gameplay experience.

The project demonstrates three key stages of game design:

1. Researching an established game system and understanding its underlying mechanics.
2. Adapting those mechanics into a different genre while introducing original gameplay features.
3. Evaluating the resulting systems through technical implementation, probability testing and economy balancing.

End of Document